



WeCloudData

RAG Evaluation

RAG





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Evaluating RAG Systems

RAG Evaluation



Why Do We Need Evaluation?

RAG Evaluation

Imagine Two Chatbots...

Chatbot A



Answers quickly.



But sometimes
invents information.



Q: What is the tallest
mountain in the world?



A: Mount Everest is actually
part of the Alps.



VS

Chatbot B



Retrieves the correct
documents.



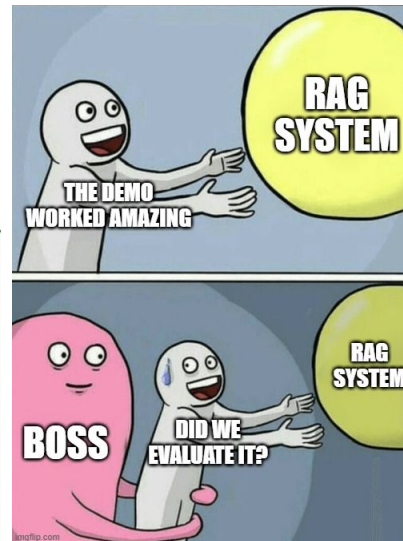
Gives trustworthy
answers.



Q: What is the tallest
mountain in the world?



A: Mount Everest is the tallest
mountain in the world, with
an elevation of 8,848.86 meters
above sea level.



A good demo does not guarantee a reliable system.



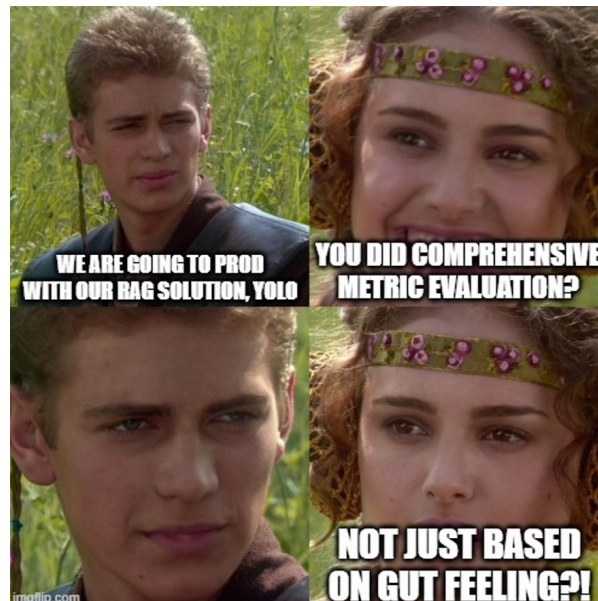
What is RAG Evaluation?

RAG Evaluation

RAG Evaluation measures **how well a Retrieval-Augmented Generation system performs.**

It helps answer questions like:

- Did we retrieve the right documents?
- Did the model answer correctly?
- Was the answer grounded in the retrieved information?



<https://www.recordlydata.com/blog/evaluating-embedding-and-reranker-models-for-finnish>



Why and When do we need RAG Evaluation?

RAG Evaluation

Why?

Without evaluation, we don't know whether mistakes come from

- Retrieval
- Generation
- Both

When?

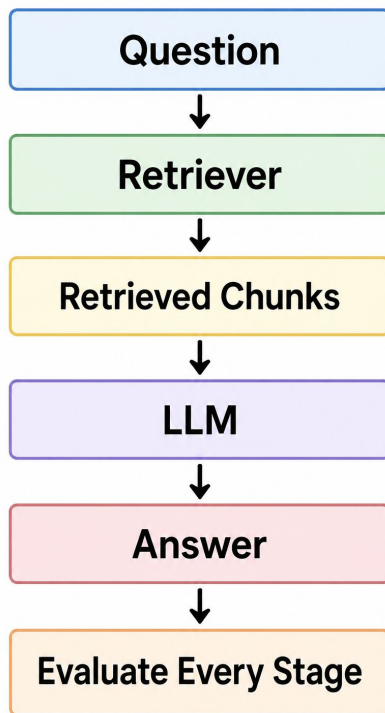
Evaluate

- Before deployment
- After changing embeddings
- After changing prompts
- After changing chunking
- Continuously in production



RAG Evaluation

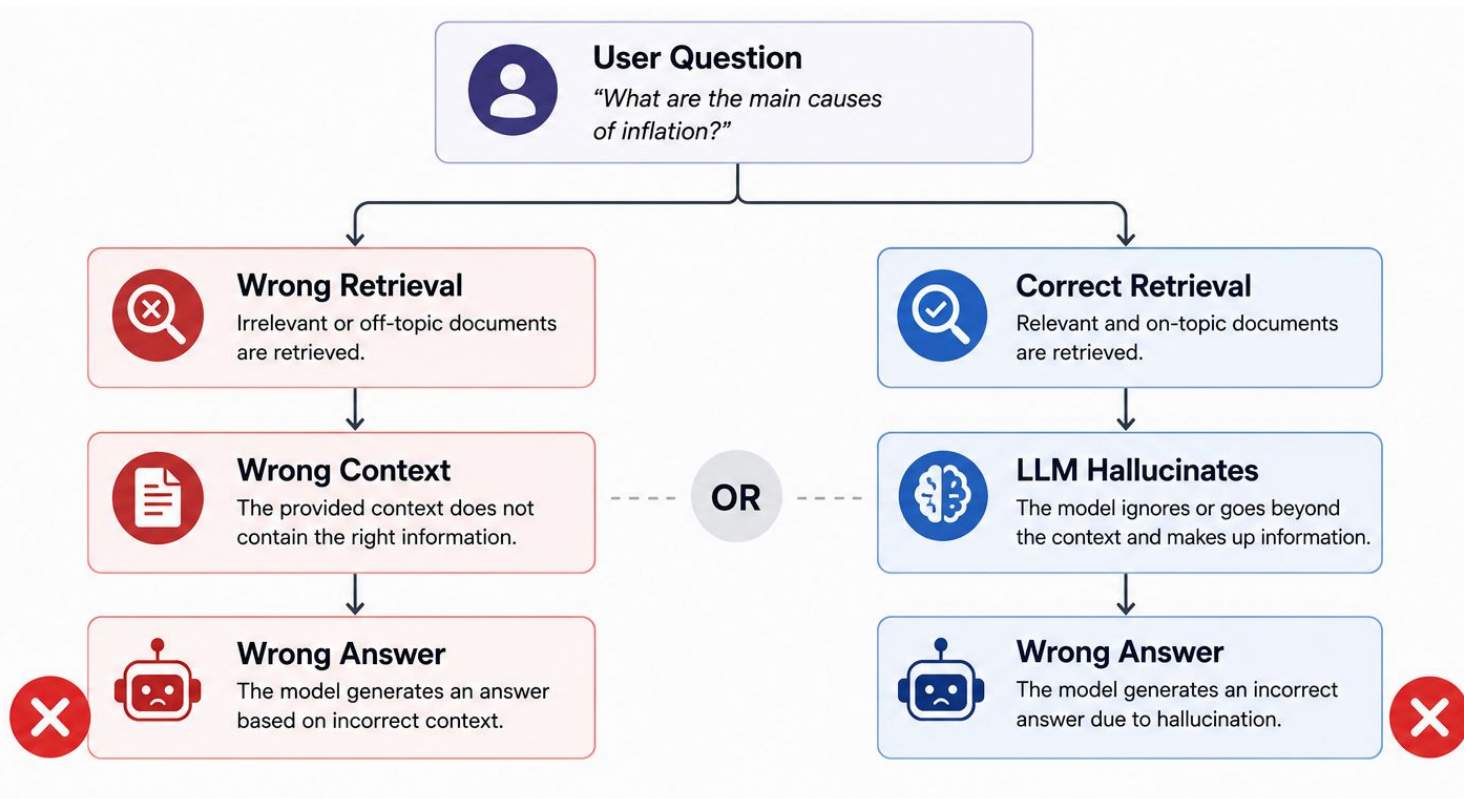
RAG Evaluation





What can go wrong in RAG Systems?

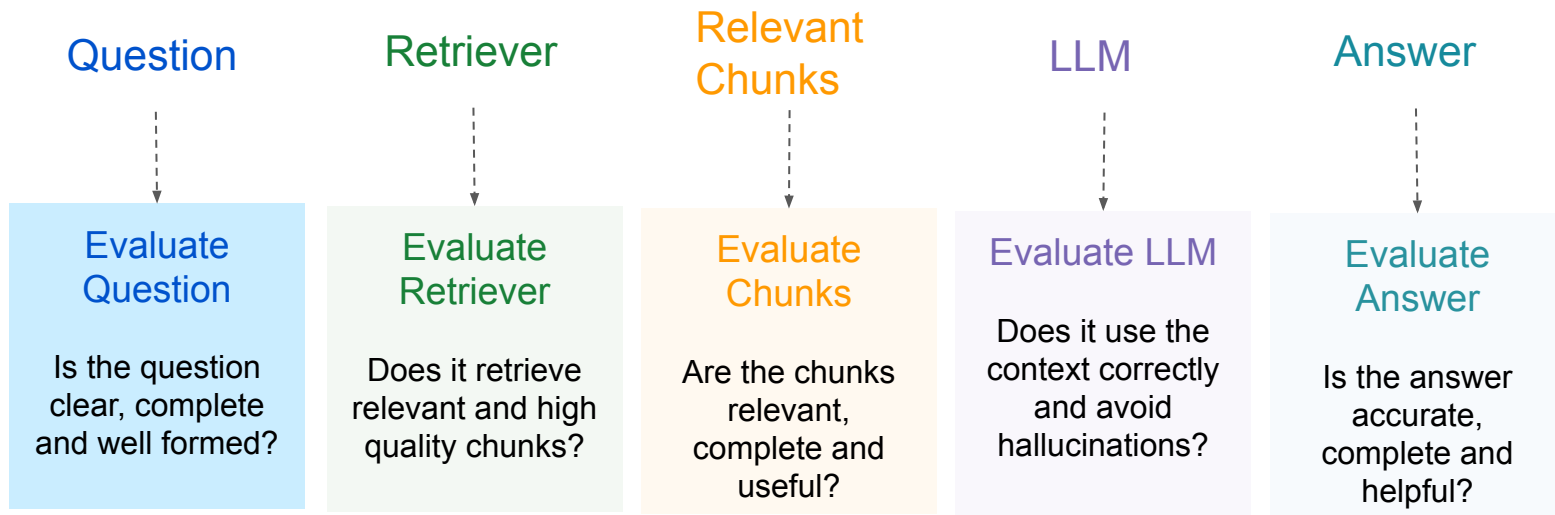
RAG Evaluation





What should we Evaluate?

RAG Evaluation





Evaluating Retrieval

RAG Evaluation

What?

Did the retriever find the correct chunks?

Questions to Ask

- Were relevant chunks retrieved?
- Were irrelevant chunks avoided?
- Were enough chunks retrieved?



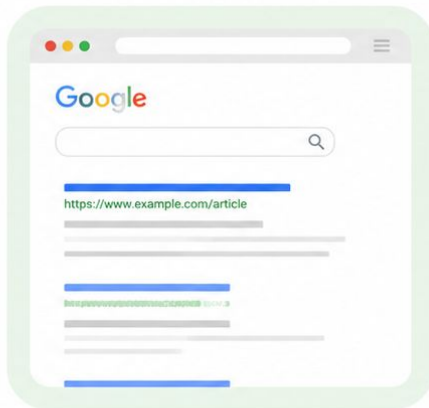
Evaluating Retrieval Analogy

RAG Evaluation

Searching Google

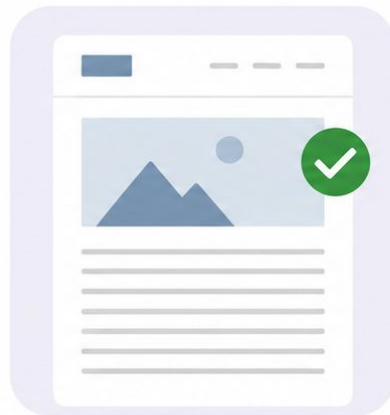


Finding right webpage



≠

Reading it correctly



Getting information is easy, understanding it correctly is the challenge.



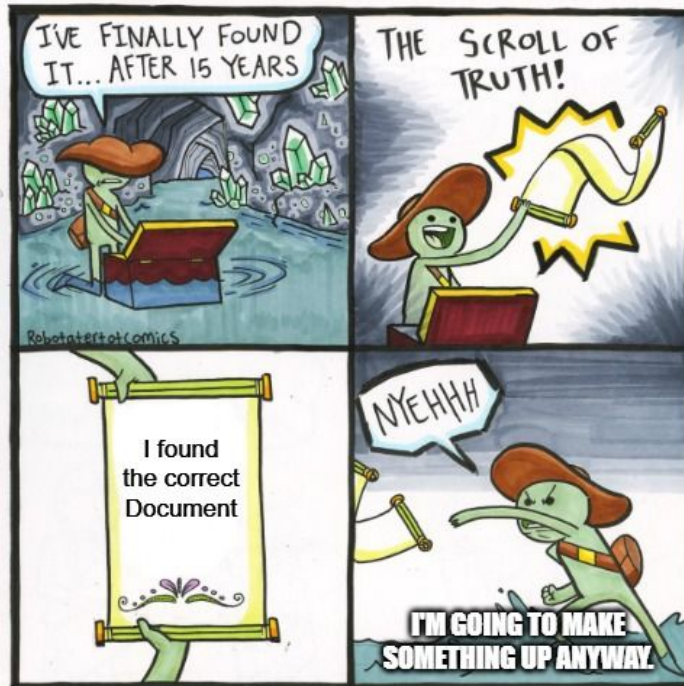
Evaluating Generation

RAG Evaluation

Now assume retrieval was perfect.

Did the LLM

- Answer correctly?
- Use retrieved context?
- Add unsupported information?

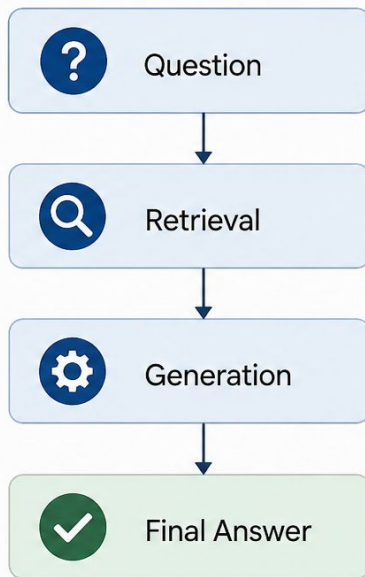




End-to-End Evaluation

RAG Evaluation

Finally evaluate the entire system



	Sometimes
Retriever	★★★★★
Generator	★★☆☆☆
Overall	★★★☆☆



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Common Evaluation Metrics

RAG Evaluation



Retrieval Metrics

RAG Evaluation

Precision

Recall

MRR



Precision

RAG Evaluation

Precision tells How many retrieved chunks were actually relevant?



Friend recommends
10 restaurants.



Only 2 are good.



Precision is low.

10 Recommended Restaurants



Good



Good



Not Good



Not Good



Not Good



Not Good



Not Good



Not Good



Not Good



Not Good



Not Good

$$\text{Precision} = \frac{\text{Number of Good Recommendations}}{\text{Total Recommendations}} = \frac{2}{10} = \mathbf{0.20}$$



Recall

RAG Evaluation

Recall tells Did we retrieve **all** the relevant information?



Open-book exam.



You found **one** chapter.

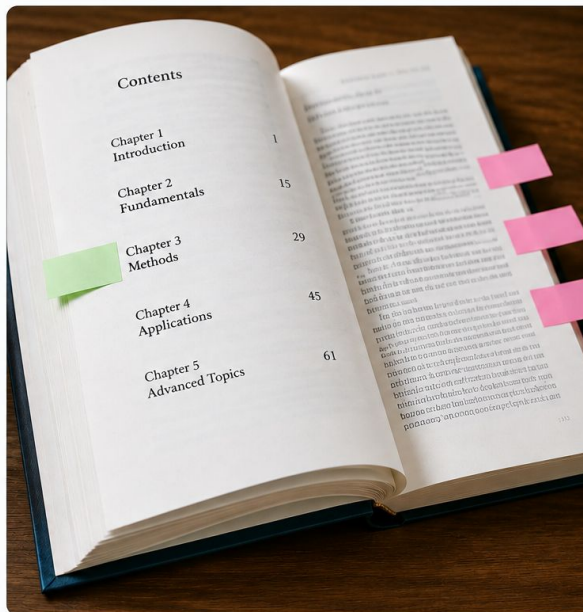


Missed **three** others.

What is Recall?

Recall measures how many of the relevant items you were able to find.

$$\text{Recall} = \frac{\text{Relevant items retrieved}}{\text{Total relevant items}}$$





MRR(Mean Reciprocal Rank)

RAG Evaluation

MRR tells us how early the first relevant result appears.

Query 1

Question

- 1 ✓ Correct Result
- 2 Wrong Result
- 3 Wrong Result

...

Correct result at R=1
RR = 1.00

Query 2

Question

- 1 Wrong Result ✗
- 2 ✓ Correct Result
- 3 Wrong Result

...

Correct result at R=2
RR = 0.50

Query 3

Question

- 1 Wrong Result ✗
- 2 Wrong Result ✗
- 3 Wrong Result ✗
- 4 ✓ Correct Result

...

Correct result at R = 4
RR = 0.25

Average RR → MRR = 0.58



Generation Metrics

RAG Evaluation

Faithfulness

Answer Correctness

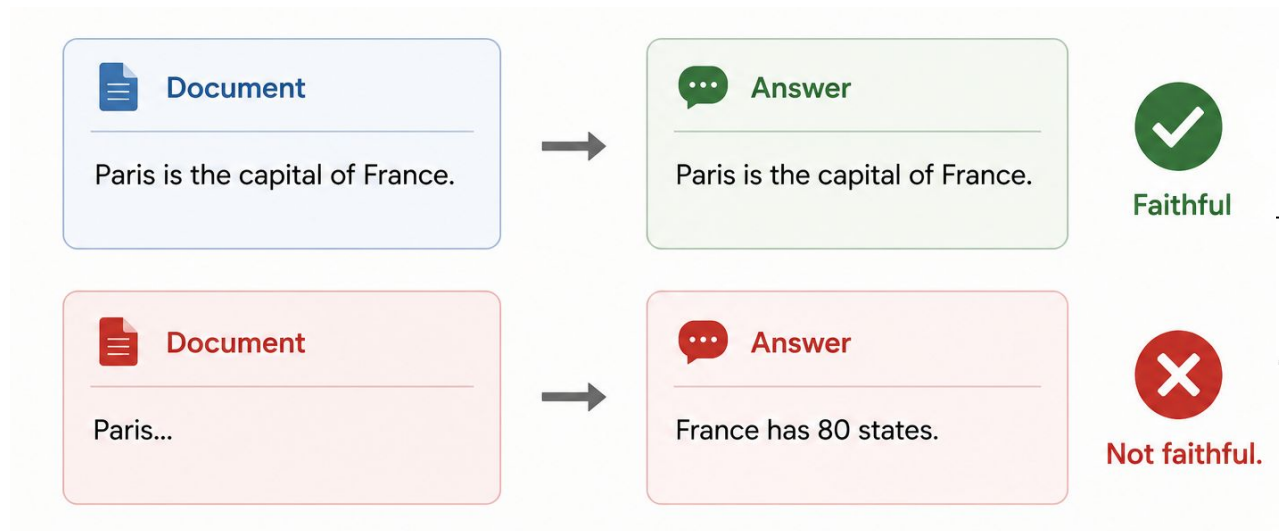
Relevance



Faithfulness

RAG Evaluation

Faithfulness tells Did the answer stay faithful to the retrieved documents?



Document:
I never
said that



LLM: But it
sounded right



The answer should not introduce unsupported facts.



Answer Relevance

RAG Evaluation

Answer relevance measure: How well the generated answer actually addresses the user's question?

Example	But Irrelevant
 Question Who wrote Romeo and Juliet?  Generated Answer William Shakespeare was a famous English playwright.  Correct The statement is factually correct.	 Question Who wrote Romeo and Juliet?  Generated Answer William Shakespeare was a famous English playwright.  Irrelevant It does not answer the question. No information about authorship.



The answer is correct but irrelevant to the question:

Low relevance = Poor Quality



Answer Correctness

RAG Evaluation

Answer correctness measures how well the generated answer matches the ground truth or reference answer.

Example 1: Correct Answer

User Question : What is the capital of Japan?

Ground Truth : Tokyo

Generated Answer : Tokyo

Answer Correctness: High

Example 2: Incorrect Answer

User Question : What is the capital of Japan?

Ground Truth : Tokyo

Generated Answer : Kyoto

Answer Correctness: Low

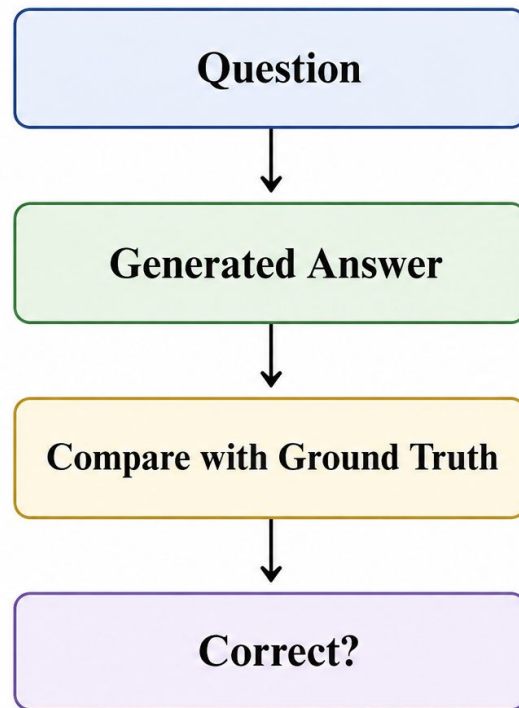


Why Does Answer Correctness Matter?

RAG Evaluation

A RAG system should not only retrieve relevant information, it should also produce **correct answers**.

High answer correctness means users can trust the system's responses.





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RAG Evaluation Framework

RAG Evaluation



Manual Vs Automated Evaluation

RAG Evaluation

Manual	Automated
Human Judges	Evaluation Models
Slow	Fast
Expensive	Scalable



Popular RAG Evaluation Frameworks

RAG Evaluation



ragas

<https://github.com/vibrantlabsai/ragas>

DeepEval.

<https://www.zdoc.app/en/confident-ai/deepeval>



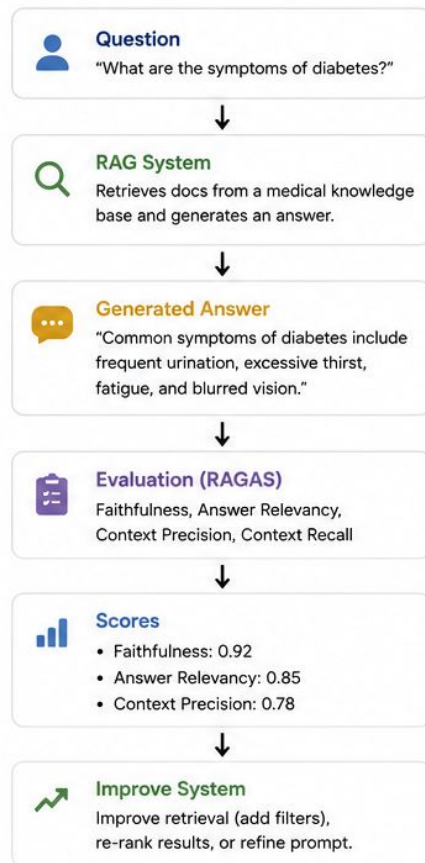
LangSmith

<https://blogs.utkarshraiput.com/langsmith-part-33-of-generative-ai-for-js-developers>



Evaluation Workflow Example

RAG Evaluation





Evaluation Tools Comparison

RAG Evaluation

	RAGAS	DeepEval	LangSmith
Focus	RAG specific metrics	LLM & RAG evaluation	Observability & tracing
Key Metrics	Faithfulness, answer relevance, context relevance	Correctness, faithfulness, relevance, safety metrics	Latency, cost, traces, token usage, evaluation scores
Strengths	Deep RAG evaluation	Easy to use, CI/CD-friendly	Debugging, monitoring, and production visibility
Best For	Evaluating retrieval quality and answer grounding	Evaluating LLM outputs and building eval suites	Monitoring and observing LLM applications
Example Use Case	Check whether answers are grounded and contexts are relevant	Evaluate correctness, tone, and safety	Trace performance, debug prompts, and monitor user interactions



Continuous Evaluation

RAG Evaluation





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Improving RAG Systems

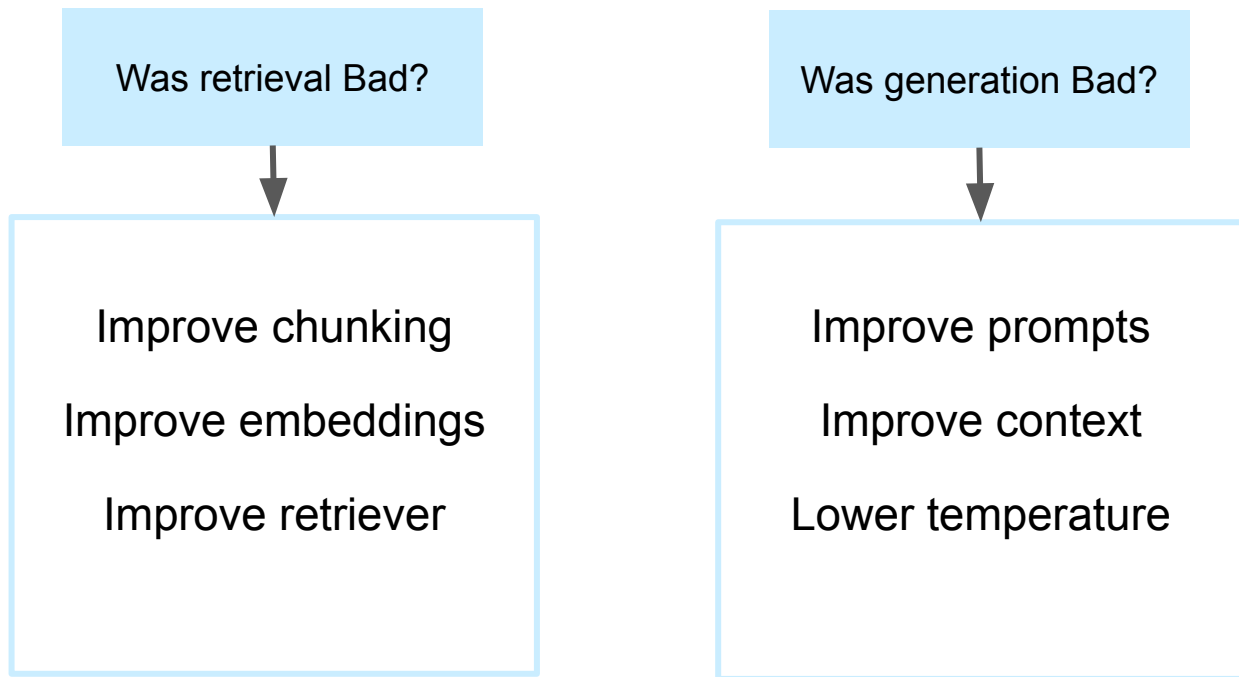
RAG Evaluation



Improving RAG Systems

RAG Evaluation

When scores are **low** ask:





What should you tune?

RAG Evaluation

Low Score	Possible Cause	Possible Fix
Low Precision	Too many irrelevant chunks	Filtering, reranking, adjust top-k
Low Recall	Important context is missing	Improve chunking, query rewriting, retrieve more
Low Faithfulness	Unsupported claims	Grounding prompt, citations, allow “I don’t know”
Low Relevance	Answer does not address the question	Clarify prompt and output requirements



Summary

- Why RAG systems must be evaluated.
- Difference between retrieval and generation evaluation.
- Common retrieval metrics (Precision, Recall, MRR).
- Common generation metrics (Faithfulness, Answer Relevance, Answer Correctness).
- Manual vs automated evaluation.
- Popular evaluation frameworks.
- Continuous improvement cycle.



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Thank You

We'd love to hear from you. If you have any questions,
feel free to reach out to our team anytime!